There is one spin 1 particle of mass μ for each eigenvalue μ^2 of the matrix (27.4.36).

Putting this all together, we see that for each eigenvalue m^2 of the matrix $\mathscr{M}^*\mathscr{M}$ there are two self-charge-conjugate spinless particles and one Majorana fermion of mass m; for each non-zero eigenvalue of the matrix μ^2_{AB} there are one self-charge-conjugate spinless boson, two Majorana fermions, and one self-charge-conjugate spin 1 boson of mass μ; and for each zero eigenvalue of this matrix there is one Majorana fermion and one self-charge-conjugate spin 1 boson of zero mass. It is not surprising that the particle multiplets for each zero or non-zero mass are just the same as we found by direct use of the supersymmetry algebra in Sections 25.4 and 25.5. What *is* a bit surprising is that the masses of the gauge and chiral particles are unaffected by each other. The masses m that are given by eigenvalues of $(\mathscr{M}^*\mathscr{M})_{nm}$ and the particles with these masses are just what they would be in a theory of chiral superfields without gauge superfields, and the masses μ that are given by eigenvalues of μ^2_{AB} and the particles with these masses are just what they would be in a theory of gauge superfields with no chiral superfields.

For future use in Section 27.9 we will now apply the method described in Section 26.7 to construct the supersymmetry current for the supersymmetric gauge Lagrangian (27.4.1). In the gauge used earlier, an infinitesimal supersymmetry transformation changes V_A, λ_a, and D_A by the amounts (27.3.4)–(27.3.6). Adding the Noether supersymmetry current given by Eq. (26.7.2) for these fields to the Noether current for ϕ_n, ψ_n, and $\mathscr{F}_n$ already given in Eq. (26.7.8), with derivatives replaced by gauge-invariant derivatives, gives the total Noether supersymmetry current:

$$\begin{aligned} N^\mu = & \sum_A f_A^{\mu\nu}\gamma_\nu\lambda_A - \frac{1}{8}\sum_A f_{A\rho\sigma}[\gamma^\rho,\gamma^\sigma]\gamma^\mu\lambda_A - \frac{1}{2}i\sum_A D_A\gamma_5\gamma^\mu\lambda_A \\ & +\frac{1}{\sqrt{2}}\sum_n\Big[2(D^\mu\phi)^*_n\psi_{nL} + 2(D^\mu\phi)_n\psi_{nR} + (\not{D}\phi)_n\gamma^\mu\psi_{nR} \\ & +(\not{D}\phi)^*_n\gamma^\mu\psi_{nL} - \mathscr{F}_n\gamma^\mu\psi_{nR} - \mathscr{F}^*_n\gamma^\mu\psi_{nL}\Big]\,. \end{aligned} \qquad (27.4.37)$$

This is not the supersymmetry current, because the Lagrangian density is not invariant under supersymmetry; instead, its change is the derivative

$$\delta\mathscr{L} = \partial_\mu\Big(\bar{\alpha}K^\mu\Big)\,, \qquad (27.4.38)$$